

In your genes

Genetic algorithm brings evolutionary practices to programming

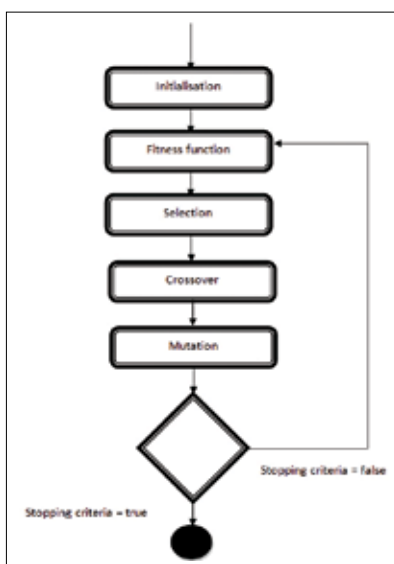
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With the advent of Machine Learning, Artificial Intelligence and Neural Networks a particular type of algorithm, Genetic Algorithms, have been used heavily. That is exactly what we are going to explore today and see how it can be used to build a game very similar to 'Space Invaders'.

In the field of Data Science, Genetic Algorithm is a heuristic method that emulates the Darwinian process of natural selection among a population where the fittest individuals are selected and are allowed to cross between themselves and mutate to produce better performing individuals, essentially bettering the entire population. It belongs to the superset of the evolutionary algorithms. It can also be thought of an optimization or a search technique.

ABOUT THE GAME

The invaders are going to multiply in every five second and our objective should be to keep destroying them until none exists. The invaders learn from our behavior and keep developing immunity against our attacks. The game begins with 4 invaders which is also the minimum number and can get to a maximum of 100. It can thus be inferred that we can never win the game and if the number of invaders reach a hundred, we lose. To rephrase,



The process

the objective should be to keep the invaders to number as few as possible on the screen. There are a few biological terms used in this context but they are easily relatable. Each invader has 4 genes (characteristics or attributes): speed, velocity, probability of its changing direction, size. At every 5 seconds, a few invaders are chosen from their entire lot by a measure of fitness following which they mate and produce children (which are supposedly better performing than their predecessors). A flowchart of the process helps for a better visualization as shown above.

We begin by initializing a population of solutions by a fitness value that is computed from a function. We select the individuals higher up in the ranks of their fitness (which may just

as well be a numerical value). Thereafter comes the crossover stage where they are then made to reproduce which leads to offspring which have qualities inherited from their parent. To give them variety, we allow them to mutate where a single individual characteristic may be altered by a bit. This procedure is carried on in a loop until the stopping criteria has been satisfied. In this case, the stopping criterion is reached if the number of invaders reaches its maximum. Let's understand the code bit by bit now

The lowest level of the code, `evolution.py` is where the central genetic algorithm has been implemented which is basically our goal. However, we shall ride through the rest of the code to see how the structure of this game can be implemented in Python.

We will use Python as our programming language for the generation of the code. There shall be four modules which we shall import, namely, `sge`, `game`, `objects` and `os`. The `sge` is a wrapper around `pygame`, which is one of the famous game building libraries in Python. The game object defines the essentials for the game logic. The objects stand for the players and the invaders and the `os` shall provide the operating system.

THE IMPLEMENTATION

We begin by setting the path to the repository of the game and then initialize the game object specific parameters like the wall height and width, the color of the screen, resolution as a background layer and set the



The player and the invaders